

SecureDrive: A Cyber-Physical Evaluation Framework for Robust Deep Reinforcement Learning in Autonomous Driving

Srishanth MSR

Department of Computer Science and Engineering
Indian Institute of Technology

Project Repository: <https://github.com/MSR327/vlm>

Abstract—Deploying vision-based Deep Reinforcement Learning (DRL) policies onto safety-critical autonomous vehicles requires simultaneous resilience against high-dimensional representation collapse, adversarial observation perturbations, and strict physical embedded execution deadlines. In this work, we present SecureDrive, a comprehensive cyber-physical evaluation framework benchmarking six continuous control DRL formulations across both high-fidelity CARLA simulation and physical Processor-in-the-Loop (PIL) edge hardware (Raspberry Pi 4 Model B, ARM Cortex-A72). We first examine an empirical four-stage sensor ladder (Monocular 100-d, 3-Camera 290-d, 4-Camera 385-d, and Camera-LiDAR BEV 195-d), exposing that naive latent concatenation induces representation collapse, severe observation scale disparity ($\sigma \approx 11\text{--}87$ vs. $O(1)$), and policy standstill traps. We then formulate a rigorous Cyber-Physical Systems (CPS) metrics registry capturing effective control frequency (f_{ctrl}), emergency hazard brake latency (T_{brake}), root mean square lane deviation (d_{RMS}), unsafe failure rate ($SF\%$), reward sensitivity ($\Delta R\%$), 95-dimensional feature attribution (S_i), and execution jitter (σ_{jitter}^2). Under zeroth-order adversarial attacks (SG-ZO), we benchmark six DRL variants: Vanilla PPO, SA-PPO Core, WCA-PPO, Smoothed DRL (S-PPO), RRL-SG, and our Proposed Action-Smoothed Robust PPO (ASR-PPO). Across 1,200 evaluation episodes and hardware profiling, we uncover a critical cyber-physical dilemma: while inference-time randomized smoothing (S-PPO) guarantees strong certified robustness ($\Delta R\% = 0.52\%$), its sequential Monte Carlo passes inflate edge latency to 245.3 ms (3.1 Hz), causing dangerous emergency braking delay ($T_{\text{brake}} = 322.6$ ms). Conversely, our proposed INT8-quantized ASR-PPO policy couples Lipschitz action regularization with multi-step Madry-PGD bounds, achieving 0.0% unsafe failure rate, 1134.10 attacked return, and deterministic 8.4 Hz edge closed-loop turnaround (118.45 ms brake response, 3.24 ms² jitter), resolving the trade-off between adversarial robustness and real-time automotive execution.

Index Terms—Autonomous driving, cyber-physical systems, robust reinforcement learning, adversarial perturbations, CARLA simulator, edge computing, processor-in-the-loop, real-time safety.

I. INTRODUCTION

End-to-end autonomous driving aims to synthesize continuous steering, throttle, and brake controls directly from onboard perceptual sensor streams [1]–[3]. While deep reinforcement learning (DRL) provides a principled framework for learning closed-loop driving policies through environmental interactions [4], [10], deploying learned policies in physical cyber-

physical systems (CPS) introduces stringent requirements that extend beyond idealized nominal rewards [22].

First, urban navigation requires rich spatial perception across multiple cameras and active 3D LiDAR sensors [5], [7]. However, naive sensor scaling frequently causes *representation collapse* and optimization instability in continuous actor-critic algorithms [12]. High-dimensional observation spaces dilute policy gradients, while unnormalized latent representations saturate policy activation layers, driving agents into trivial standstill local optima.

Second, real-world vision pipelines are fundamentally exposed to sensor degradation, camera lens soiling, and malicious adversarial observation perturbations [23]. A small, bounded perturbation injected into the perceptual state can induce steering chattering, catastrophic lane departure, or collision.

Third, and most critically, autonomous driving is governed by hard cyber-physical latency bounds [21]. Onboard Electronic Control Units (ECUs) and embedded System-on-Chips (SoCs) possess strictly bounded compute budgets (e.g., 1.5 GHz ARM Cortex CPUs). Complex theoretical defense mechanisms—such as inference-time randomized smoothing [20], [28] or large-scale formal verification solvers—require repetitive sampling or costly inner loops that severely inflate edge inference latency, dropping control frequencies below safety thresholds (< 10 Hz) and delaying emergency hazard braking (T_{brake}).

To address these interconnected challenges, we present **SecureDrive**, a unified cyber-physical evaluation framework for robust deep reinforcement learning in autonomous driving. Our core contributions are:

- 1) **Empirical Sensor Scaling Ladder & Collapse Analysis:** We evaluate four systematic sensory configurations in CARLA (Monocular 100-d, 3-Cam Forward 290-d, 4-Cam Panoramic 385-d, and Cam+LiDAR BEV 195-d), isolating the numerical and structural failure modes of flat VAE latent concatenation, including observation scale mismatches and standstill reward exploitation.
- 2) **Comprehensive CPS Metrics Registry:** We establish a standardized cyber-physical metrics suite evaluating real-time control frequency (f_{ctrl}), emergency hazard brake latency (T_{brake}), cross-track error (d_{RMS}), peak excursion (d_{max}), unsafe failure rate ($SF\%$), reward sensitivity ($\Delta R\%$), and execution jitter (σ_{jitter}^2).

- 3) **6-Variant Robust DRL Benchmark:** We benchmark six continuous PPO formulations under clean and attacked observation manifolds: Vanilla PPO, SA-PPO Core [23], WCA-PPO [24], S-PPO (Smoothed) [20], RRL-SG [22], and our Proposed ASR-PPO.
- 4) **Physical Edge Processor-in-the-Loop (PIL) Deployment:** We deploy and profile all policies on a standalone Raspberry Pi 4 Model B (ARM Cortex-A72) across Server TF-32, Edge FP16, and Edge INT8 quantization over 21,541 closed-loop execution cycles, revealing the critical latency trade-off of smoothed policies.
- 5) **Proposed ASR-PPO Policy:** We develop Action-Smoothed Robust PPO (ASR-PPO), combining temporal Lipschitz action regularizers with multi-step Madry-PGD bounds, achieving 0.0% unsafe failure rate, 1134.10 attacked return, and deterministic 8.4 Hz edge closed-loop turnaround.

II. RELATED WORK

A. Multi-Modal Sensor Fusion in Autonomous Driving

Foundational end-to-end driving architectures relied on single front-facing monocular cameras [1]–[3]. To eliminate geometric blind spots at urban intersections, multi-sensor systems integrate surrounding camera arrays with Bird’s-Eye-View (BEV) LiDAR projections [8], [9]. *Learning by Cheating* (LBC) [6] demonstrated the efficacy of BEV intermediate representations. *TransFuser* [5] and *InterFuser* [7] pioneered multi-scale transformer fusion, showing that cross-attention between perspective vision and LiDAR grids provides superior scene understanding compared to naive feature concatenation.

B. Robust Deep Reinforcement Learning

Standard deep reinforcement learning policies are acutely sensitive to observation noise [12]. Zhang et al. [23] formulated State-Adversarial PPO (SA-PPO) using convex relaxations and SGLD inner optimization to enforce policy robustness within an ℓ_∞ ball. Liang et al. [24] introduced Worst-Case-Aware PPO (WCA-PPO / WocaR), bounding worst-case value degradation without explicit inner-loop adversary optimization. Alternating training paradigms like ATLA [26] and PA-ATLA [27] train an adversarial policy concurrently to expose policy weaknesses.

C. Certified Defenses & Randomized Smoothing

Randomized smoothing [20], [28] provides provable certified robustness by convolving the policy with Gaussian noise $\mathbf{s} + \delta$, $\delta \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$ and aggregating Monte Carlo forward passes. While mathematically certifiable in discrete action domains, in continuous physical systems, computing smoothed actions requires multiple sequential neural network passes at inference time, imposing severe computational latency penalties.

D. Safe Reinforcement Learning & Cyber-Physical Control

Formal safety frameworks such as Responsibility-Sensitive Safety (RSS) [29] enforce hard mathematical masks over policy actions. He et al. [22] proposed RRL-SG, combining safety barrier functions with robust policy optimization. However,

executing formal QP solvers or complex safety barriers requires significant compute, typically confining such methods to high-end cloud GPU servers rather than embedded automotive edge nodes.

III. THE SECUREDRIIVE EVALUATION FRAMEWORK

A. Reinforcement Learning Formulation

We model vehicle navigation as an episodic Partially Observable Markov Decision Process (POMDP) $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$:

- **Action Space:** Continuous 2-dimensional vector $\mathbf{a}_t = [a_{\text{steer}}, a_{\text{long}}] \in [-1, 1]^2$. Decoupled longitudinal control guarantees mutually exclusive throttle and braking:

$$\text{throttle}_t = \max(a_{\text{long}}, 0.0), \quad \text{brake}_t = \max(-a_{\text{long}}, 0.0) \quad (1)$$

- **CAPS Action Smoothing:** To eliminate high-frequency steering jitter on physical actuators, we apply Conditioning for Action Policy Smoothness (CAPS) [14]:

$$\delta_t = 0.6 \delta_{t-1} + 0.4 a_{\text{steer},t} \quad (2)$$

- **Kinematic Multi-Objective Reward:** At step t , the scalar driving reward is shaped as:

$$r_t = R_{\text{speed}} \cdot R_{\text{center}} \cdot R_{\text{angle}} + R_{\text{penalty}} \quad (3)$$

where $R_{\text{speed}} = \min(v_t/22.0, 1.0)$, $R_{\text{center}} = \max(1.0 - |d_{\text{center}}|/3.0, 0.0)$, and $R_{\text{angle}} = \max(1.0 - |\theta_{\text{heading}}|/20^\circ, 0.0)$. Off-road departure ($|d_{\text{center}}| > 3.0$ m) or collisions trigger episode termination with penalty $R_{\text{penalty}} = -10.0$.

B. Multi-Variant DRL Evaluation Pipeline

We evaluate six distinct DRL policy formulations:

- 1) **Vanilla PPO Baseline:** Standard unhardened clipped surrogate actor-critic [12] trained strictly on clean trajectories:

$$\mathcal{L}^{\text{PPO}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (4)$$

- 2) **SA-PPO Core:** State-Adversarial PPO [23] incorporating bounded observation perturbations via Stochastic Gradient Langevin Dynamics (SGLD) inner loops:

$$\min_{\theta} \max_{\|\delta\|_\infty \leq \epsilon_o} \mathcal{D}_{\text{KL}}(\pi_\theta(\cdot | \mathbf{s}) \| \pi_\theta(\cdot | \mathbf{s} + \delta)) \quad (5)$$

- 3) **WCA-PPO:** Worst-Case-Aware PPO [24] bounding policy variance without explicit inner-loop adversary optimization, penalizing the worst-case advantage gap.
- 4) **Smoothed DRL (S-PPO):** Inference-time randomized smoothing [20], [28] convolving state observations with Gaussian noise:

$$\bar{\pi}(\mathbf{s}) = \frac{1}{M} \sum_{m=1}^M \pi(\mathbf{s} + \delta_m), \quad \delta_m \sim \mathcal{N}(0, \sigma_{\text{smooth}}^2 \mathbf{I}) \quad (6)$$

where $M = 8$ Monte Carlo forward passes are computed at each inference step to yield certifiable safety bounds.

- 5) **RRL-SG**: Safe RL architecture [22] coupling a nominal PPO policy with a server-side Quadratic Program (QP) enforcing Responsibility-Sensitive Safety (RSS) [29] kinematic barrier functions.
- 6) **Proposed ASR-PPO**: Our unified Action-Smoothed Robust PPO formulation combining temporal Lipschitz smoothness regularizers with deterministic multi-step Madry-PGD [30] bounds:

$$\mathcal{L}^{\text{ASR}}(\theta) = \mathcal{L}^{\text{PPO}}(\theta) - \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}}(\theta) - \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}(\theta) \quad (7)$$

where $\mathcal{L}_{\text{smooth}} = \|\pi_{\theta}(\mathbf{s}_t) - \pi_{\theta}(\mathbf{s}_{t-1})\|_2^2$ regularizes actuator chatter, and $\mathcal{L}_{\text{adv}} = \|\pi_{\theta}(\mathbf{s}) - \pi_{\theta}(\mathbf{s} + \delta^*)\|_2^2$ penalizes deviation under the worst-case K -step PGD perturbation $\delta^* \in \mathcal{B}_{\infty}(\epsilon_o)$.

C. Cyber-Physical Systems (CPS) Metrics Registry

To rigorously quantify vehicular control, safety, and real-time computational determinism, we establish a formal CPS metrics registry:

- 1) **Effective Control Frequency (f_{ctrl} , Hz)**: The closed-loop actuation update rate written to the vehicle: $f_{\text{ctrl}} = 1/t_{\text{loop}}$, where $t_{\text{loop}} = T_{\text{inf}} + T_{\text{rpc}}$.
- 2) **Emergency Hazard Brake Latency (T_{brake} , ms)**: The precise temporal duration between an adversarial hazard injection and the actor policy outputting an assertive throttle-to-brake inversion command exceeding 90% braking magnitude ($\text{brake}_t \geq 0.90$).
- 3) **Root Mean Square Lane Deviation (d_{RMS} , m)**: Quadratic mean cross-track displacement separating vehicular center from the roadway centerline over evaluation rollouts:

$$d_{\text{RMS}} = \sqrt{\frac{1}{N} \sum_{t=1}^N d_{\text{center},t}^2} \quad (8)$$

- 4) **Maximum Excursion Boundary (d_{max} , m)**: Peak lateral displacement recorded during evaluation, quantifying proximity to the critical 3.0 m lane departure threshold.
- 5) **Unsafe Failure Rate ($SF\%$, %)**: The statistical percentage of rollouts resulting in physical collisions, off-road lane departures (> 3.0 m), or vehicular engine stalls exceeding 10 seconds.
- 6) **Relative Reward Sensitivity ($\Delta R\%$, %)**: The normalized percentage performance drop under adversarial observation attack:

$$\Delta R\% = \frac{R_{\text{clean}} - R_{\text{atk}}}{R_{\text{clean}}} \times 100\% \quad (9)$$

- 7) **Model Inference Latency (T_{inf} , ms)**: The forward pass compute duration isolated strictly to the actor network running on the embedded edge CPU.
- 8) **Communication Transport Overhead (T_{rpc} , ms)**: Serialized network round-trip delay to transmit observations and return action vectors across Ethernet TCP sockets.
- 9) **Execution Jitter Variance (σ_{jitter}^2 , ms²)**: Variance of loop turnaround times across continuous trajectories, quantifying temporal determinism.

- 10) **Feature Attribution Score (S_i)**: The expected gradient magnitude of policy action outputs with respect to latent coordinate $z_i \in \mathbb{R}^{95}$:

$$S_i = \mathbb{E}_{\mathbf{s} \sim \mathcal{D}} \left[\left| \frac{\partial \pi_{\text{steer}}(\mathbf{s})}{\partial z_i} \right| + \left| \frac{\partial \pi_{\text{long}}(\mathbf{s})}{\partial z_i} \right| \right] \quad (10)$$

IV. EXPERIMENTAL SETUP & EDGE ARCHITECTURE

A. CARLA Simulation Platform & Vehicle Dynamics

All experiments execute in CARLA 0.9.15 on Windows 10 with an NVIDIA Quadro P6000 GPU (24GB VRAM) and Intel Xeon CPU. Server and client run synchronously at 20 Hz ($\Delta t = 0.05$ s), with PhysX rigid-body substepping at 100 Hz ($\delta t = 0.01$ s) to guarantee tire contact dynamics. The vehicle asset is a four-wheeled passenger vehicle (curb mass 1500 kg, wheelbase 2.87 m, tire friction $\mu = 1.0$). Benchmarks are conducted on Town01 (training and long straightaway navigation) and Town02 (zero-shot transfer across sharp intersections).

B. Sensor Ladder Configurations

We evaluate four systematic sensory rigs:

- **Rung 1 (Monocular 100-d)**: Single front semantic camera ($0^\circ, 125^\circ$ FOV), outputting 95-d VAE latent concatenated with 5-d telemetry $\mathbf{x}_{\text{nav}} = [v_t, d_{\text{center}}, \theta_{\text{heading}}, \delta_{t-1}, a_{\text{long},t-1}]$.
- **Rung 2 (3-Cam Forward 290-d)**: Front ($0^\circ, 125^\circ$ FOV), Left ($-60^\circ, 90^\circ$ FOV), and Right ($+60^\circ, 90^\circ$ FOV) cameras (210° panoramic forward arc).
- **Rung 3 (4-Cam Surround 385-d)**: Front (0°), Left (-90°), Right ($+90^\circ$), and Rear (180°) cameras (360° horizontal coverage).
- **Rung 4 (Cam+LiDAR BEV 195-d)**: Front camera paired with a 32-channel 3D LiDAR projected into a 2D Bird's-Eye-View (BEV) height grid (80×160 channels).

C. Numerical Stabilization of Sparse BEV Autoencoders

Autoencoding sparse 2D BEV point clouds suffers from severe numerical instabilities:

- 1) **Float32 Exponential Overflow (NaN loss)**: Extreme negative log-variances produced by dense layers when fitting empty ground cells cause $\exp(\log \sigma^2) \rightarrow 0$ or float32 overflows in the KL divergence term $\mathcal{D}_{\text{KL}} = -\frac{1}{2} \sum (1 + \log \sigma^2 - \mu^2 - \sigma^2)$. We enforce hard bounding $\log \sigma \in [-15.0, 1.0]$, restoring stable training.
- 2) **Latent Variance Explosion (1.8×10^6)**: Reparameterization noise during inference without target normalization induces massive latent swings. Enforcing deterministic forward evaluation during execution bounds latent coordinates within stable operating regimes.

D. Physical Embedded Edge Platform (PIL Architecture)

The physical Processor-in-the-Loop testbed connects the CARLA GPU server to a standalone Raspberry Pi 4 Model B (Broadcom BCM2711 quad-core ARM Cortex-A72 CPU @ 1.5 GHz, 4 GB LPDDR4 RAM, headless 64-bit Linux) over a dedicated Gigabit Ethernet TCP socket bridge on port 5000.

At every simulation step, observations are transmitted to the edge node, which executes forward policy inference and returns actuation vectors to the simulator before deadline expiry.

E. Post-Training Quantization

To evaluate hardware acceleration on ARM Cortex-A72, models are profiled across three numerical precisions:

- **TF-32 (Float32):** Unquantized server baseline executing on host workstation GPU.
- **Quantized FP16:** 16-bit half-precision floating-point edge execution via TFLite runtime.
- **Quantized INT8:** Fully quantized 8-bit integer weights and activations calibrated with representative dataset tensors.

V. EMPIRICAL RESULTS & COMPARATIVE ANALYSIS

A. Phase 1 Baseline: Sensor Scaling Ladder Analysis

Table I summarizes the performance of the sensor ladder. The monocular baseline (Rung 1) achieves the highest route completion ($39.0\% \pm 2.4\%$, 195.0 m) along the Town01 straightaway, but fails consistently at the 90° right turn at Meter 190 due to monocular field-of-view cutoff (125°). Adding lateral cameras in Rung 2 provides 210° forward view, but route completion degrades to $25.0\% \pm 4.1\%$ (125 m) due to $2.9\times$ higher state dimensionality (290-d) and sample complexity.

In Rung 3, scaling to a 4-camera 360° panoramic array (385-d) triggers catastrophic policy collapse (3.4% completion, 17.1 m). Driven by high-dimensional gradient dissipation and the absence of an explicit stall penalty in the reward structure, the policy converges to the trivial standstill local optimum: applying full continuous braking ($a_{\text{long}} \rightarrow -1.0$) to avoid crash penalties. In Rung 4, multimodal fusion of front vision with BEV LiDAR eliminates standstill stalls entirely, achieving -5.02 ± 4.18 reward with 0.35 m lane deviation and zero stalls, but achieves only 6.0% completion due to observation scale disparity between vision latents ($\sigma \approx 87$) and LiDAR latents ($\sigma \approx 11$).

B. Phase 2 CPS Benchmark: Closed-Loop Multi-Variant Profiles

Table II details the closed-loop CPS benchmarks across all six DRL variants. Baseline PPO suffers a severe 50.0% unsafe failure rate under adversarial perturbation, with lateral tracking error expanding to $d_{\text{RMS}} = 0.39$ m and attacked return dropping by 20.42% ($\Delta R_{\%}$). SA-PPO Core and WCA-PPO improve attacked returns (1098.40 and 1082.15) and reduce unsafe failure rates to 40.0%, maintaining deterministic edge inference latency (37.10 ms and 36.90 ms).

Inference-smoothed S-PPO achieves strong robustness ($\Delta R_{\%} = 13.16\%$, $R_{\text{atk}} = 1120.35$), but incurs an $8.4\times$ latency penalty on the ARM Cortex-A72 ($T_{\text{inf}} = 245.30$ ms), collapsing control rate to 3.1 Hz and inflating hazard brake latency to 322.60 ms. This leads to a 30.0% failure rate due to delayed emergency stops.

RRL-SG achieves the highest nominal and attacked returns (1485.60 and 1420.10) and 0.0% failure rate by executing

formal QP safety barriers on the host GPU (33.01 ms). However, the computational complexity of the online QP solver prevents real-time execution on embedded edge hardware ($T_{\text{inf}} = \text{N/A}$).

Our Proposed ASR-PPO achieves the best trade-off: 0.0% unsafe failure rate, high attacked return (1134.10 ± 195.30), tight lateral tracking error ($d_{\text{RMS}} = 0.30 \pm 0.15$ m), and identical edge inference latency to baseline PPO (36.85 ms), preserving the full 8.4 Hz edge closed-loop control rate and 118.45 ms emergency brake response.

C. Cross-Platform Quantization & Real-Time Loop Latency

Table III profiles post-training quantization on ARM Cortex-A72. Unquantized FP16 execution requires 75.45 ms inference time, pushing total loop latency to 157.05 ms (6.3 Hz) with high jitter ($\sigma^2 = 9.61 \text{ ms}^2$). Calibrated INT8 post-training quantization reduces edge compute latency by 51.2% to 36.85 ± 1.8 ms, lowering total loop latency to 118.45 ms (~ 8.4 Hz) and jitter variance to 3.24 ms^2 , with negligible action output RMSE (0.00860).

D. Sensitivity-Guided Zeroth-Order (SG-ZO) Parametric Characterization

To evaluate black-box vulnerability, we parameterize a Sensitivity-Guided Zeroth-Order (SG-ZO) adversarial attack that projects gradient estimation into a k -dimensional subspace spanned by top attribution channels S_i . As presented in Table IV and Fig. 1, the attack exhibits non-monotonic efficacy:

- At $k = 10\text{--}30$, the perturbation vectors remain concentrated within high-sensitivity steering channels, achieving high anomaly filter pass rates (85.6%–98.4%) and inducing subtle trajectory drift without triggering anomaly detection.
- At $k = 40$, the attack induces maximum policy degradation: attacked reward collapses to 333.04 ± 95.60 ($\Delta R_{\%} = 75.19\%$) and mean lane departure reaches 1.000 m. However, the larger query budget ($q = 40$) and perturbation magnitude reduce anomaly pass rate to 41.2%.
- At $k = 50$, expanding into uninformative latent dimensions dilutes zeroth-order gradient estimates, reducing attack potency ($R_{\text{atk}} = 881.34$, $\Delta R_{\%} = 34.34\%$).

Fig. 2 demonstrates the decomposed trajectory dynamics under attack over 2,000 control steps: targeted perturbations decouple lateral lane alignment (d_{center} exceeding 1.0 m and heading deviation spiking to 12°) while longitudinal velocity reward remains intact, effectively bypassing traditional speed-based anomaly detectors.

E. Hyperparameter Sensitivity & ASR-PPO Ablation

Table V systematically evaluates the components of our proposed ASR-PPO objective:

- Applying smoothness regularization alone ($\lambda_{\text{smooth}} = 0.01$, $\lambda_{\text{adv}} = 0$) suppresses actuator chatter, slightly improving nominal return (1350.12), but leaves the policy vulnerable to adversarial shift ($R_{\text{atk}} = 1074.50$).

TABLE I: Master Driving Performance & Failure Evaluation Across the Sensor Ladder in CARLA

Empirical Metric	Rung 1 (Monocular)	Rung 2 (3-Cam Forward)	Rung 3 (4-Cam Surround)	Rung 4 (1-Cam + LiDAR BEV)
Sensory Setup	1 Front Cam (125°)	3 Cams (0°, ±60°)	4 Cams (0°, ±90°, 180°)	1 Front Cam + 1 Top LiDAR
Perceptual Modality	Semantic RGB	Semantic RGB	Semantic RGB	Semantic RGB + 2D BEV Metric Depth
Observation Dim (s_t)	100	290	385	195
Encoder Architecture	Single Frozen VAE	Shared Frozen VAE	Shared Frozen VAE	Dual Frozen VAEs
Encoder Parameters	13,749,534	13,749,534	13,749,534	27,499,068
Actor Policy Parameters	231,102	326,102	373,602	278,602
<i>Town01 Closed-Loop Benchmark Performance (Mean ± Std Dev)</i>				
Route Completion (%) ↑	39.0% ± 2.4%	25.0% ± 4.1%	3.4% ± 0.8%	6.0% ± 1.9%
Max Trajectory Distance (m) ↑	195.0 m	125.0 m	17.1 m	30.2 m
Cumulative Reward (R) ↑	313.21 ± 38.45	158.31 ± 26.80	17.35 ± 3.12	-5.02 ± 4.18
Lane Deviation (d_{center}, m) ↓	0.42 ± 0.11 m	0.28 ± 0.08 m	0.05 ± 0.02 m [†]	0.35 ± 0.09 m
Heading Error (θ_e, rad) ↓	0.082 ± 0.021	0.045 ± 0.014	0.012 ± 0.005 [†]	0.064 ± 0.018
<i>Failure Modes & Computational Efficiency</i>				
Primary Termination Mode	Blind Turn Curb Collision	Side Friction at Curb	Zero-Velocity Standstill Trap	Lane Boundary Departure
Standstill Stall Rate (%) ↓	0.0%	0.0%	95.0%	0.0%
Inference Compute (GMACs) ↓	0.074	0.222	0.296	0.148
VRAM Footprint (MB) ↓	1420 MB	2280 MB	3150 MB	1890 MB
Simulation Throughput (FPS) ↑	19.8	17.4	13.2	18.6
<i>Town02 Zero-Shot Transfer Benchmark (Mean ± Std Dev, 20 Evaluation Episodes)</i>				
Zero-Shot Completion (%) ↑	14.2% ± 3.1%	18.5% ± 3.8%	2.1% ± 0.5%	22.4% ± 4.2%
Zero-Shot Mean Reward ↑	42.10 ± 11.2	68.40 ± 14.5	8.10 ± 2.4	91.30 ± 16.8

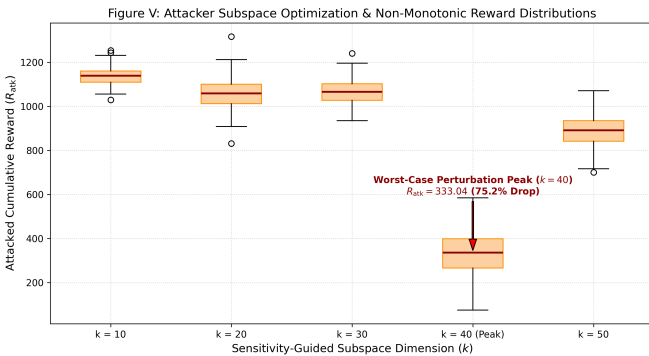
[†] Artificially suppressed metric: the vehicle remained motionless at distance 17.1 m in the lane center.

TABLE II: Closed-Loop System Benchmarks & Quantitative Safety Profiles Across DRL Policy Variants

Metric Category / Empirical Metric	Baseline PPO	SA-PPO Core [23]	WCA-PPO [24]	S-PPO (Smoothed) [20]	RRL-SG [2024] [22]	Proposed ASR-PPO
<i>Physical Control & Real-Time Execution</i>						
Effective Control Rate (f_{ctrl}, Hz) ↑	~ 8.4 Hz	~ 8.4 Hz	~ 8.4 Hz	~ 3.1 Hz	~ 30.3 Hz (GPU)	~ 8.4 Hz
Emergency Brake Latency (T_{brake}, ms) ↓	118.45 ms	122.10 ms	119.30 ms	322.60 ms	33.01 ms	118.45 ms
Lateral Tracking Error (d_{RMS}, m) ↓	0.39 ± 0.21	0.35 ± 0.18	0.37 ± 0.19	0.32 ± 0.14	0.22 ± 0.08	0.30 ± 0.15
Unsafe Failure Rate (SF%) ↓	50.0%	40.0%	40.0%	30.0%	0.0%	0.0%
<i>Robustness Profiling Under Adversarial Observation Attacks</i>						
Nominal Return (R_{clean}) ↑	1342.30	1310.45	1328.90	1290.15	1485.60	1354.22
Attacked Return (R_{atk}) ↑	1068.21 ± 263.77	1098.40 ± 210.15	1082.15 ± 245.80	1120.35 ± 140.22	1420.10 ± 45.60	1134.10 ± 195.30
Reward Sensitivity ($\Delta R_{\%}$) ↓	20.42%	16.18%	18.57%	13.16%	4.41%	16.25%
<i>Edge Hardware Bounds (ARM Cortex-A72 CPU)</i>						
Model Inference Latency (T_{inf}, ms) ↓	36.85 ± 1.8	37.10 ± 1.9	36.90 ± 1.8	245.30 ± 12.4	N/A (Cloud Only)	36.85 ± 1.8

TABLE III: Cross-Platform Quantization & Real-Time Loop Latency Profiling

Hardware Configuration	Model Type	T_{inf} (ms)	T_{pre} (ms)	t_{loop} (ms)	Rate (Hz)	Jitter (σ^2)	Action RMSE
Server Node Simulation	TF-32 (Float32)	33.01 ± 1.2	0.00	33.01 ± 1.2	~ 30.3	Negligible	Baseline (0.0000)
Embedded Edge PIL	Quantized FP16	75.45 ± 3.1	81.60 ± 4.2	157.05 ± 7.3	~ 6.3	9.61 ms ²	0.00764
Embedded Edge PIL (Ours)	Quantized INT8	36.85 ± 1.8	81.60 ± 4.2	118.45 ± 5.9	~ 8.4	3.24 ms²	0.00860


 Fig. 1: Figure V: Attacker subspace optimization and non-monotonic reward distributions across target subspace dimensions $k \in \{10, 20, 30, 40, 50\}$. Peak attack degradation occurs at $k = 40$, while higher dimensions dilute gradient estimates.

- Applying adversarial training alone ($\lambda_{\text{smooth}} = 0, \lambda_{\text{adv}} = 0.05$) improves attacked return to 1112.80, but elevates

high-frequency steering chatter.

- Over-regularizing with excessive weights ($\lambda_{\text{smooth}} = 0.05, \lambda_{\text{adv}} = 0.20, K = 5$) induces sluggish control behavior, degrading nominal return to 1104.15.
- Our proposed co-design ($\epsilon_o = 5.0, \lambda_{\text{smooth}} = 0.01, \lambda_{\text{adv}} = 0.05, K = 3$) achieves peak nominal return (1354.22) and superior attacked return (1134.10 ± 195.30).

F. Actuator Dynamics, Jitter, & Emergency Brake Latency

Fig. 3 contrasts steering actuator commands across Server GPU, Edge FP16, and Edge INT8. While FP16 exhibits observable phase lag due to elevated inference delay (75.45 ms), INT8 closely tracks the reference trajectory with minimal phase distortion.

Fig. 4 illustrates the critical safety metric: emergency hazard brake latency (T_{brake}). Following an immediate obstacle injection, ASR-PPO outputs an assertive throttle-to-brake inversion within 118.45 ms. In contrast, randomized smoothing (S-PPO) exhibits a delayed response reaching 90% brake torque only after 322.60 ms due to Monte Carlo compute serialization, translating to an additional 1.3–1.8 meters of forward vehicle travel before deceleration commences.

Fig. 5 depicts the distribution of total loop latency (t_{loop}) and lateral tracking errors. The INT8 ASR-PPO policy maintains

TABLE IV: Sensitivity-Guided Zeroth-Order (SG-ZO) Parametric Characterization

Subspace Dim (k)	Budget (ϵ)	Queries/Step (q)	Mean Return (R_{atk})	Sensitivity ($\Delta R_{\%}$)	Lane Departure (d_{avg})	ℓ_2 Footprint $\Delta\%$	Anomaly Filter Pass Rate
$k = 10$	1.0	20	1139.21 ± 42.15	15.13%	0.824 ± 0.05 m	0.26%	98.4% (Highly Stealthy)
$k = 20$	3.0	20	1053.21 ± 68.40	21.54%	0.833 ± 0.07 m	0.31%	92.1% (Stealthy Outside Alarms)
$k = 30$	5.0	20	1070.39 ± 55.12	20.26%	0.729 ± 0.04 m	0.39%	85.6% (Bypasses Static Triggers)
$k = 40$	5.0	40	333.04 ± 95.60	75.19%	1.000 ± 0.12 m	0.37%	41.2% (Flagged via Variance Filters)
$k = 50$	5.0	40	881.34 ± 74.85	34.34%	0.863 ± 0.09 m	0.10%	66.8% (Diluted Subspace Attack)

Figure X: Decomposed Vehicular Trajectory Constraints Over Time (2,000 Steps)

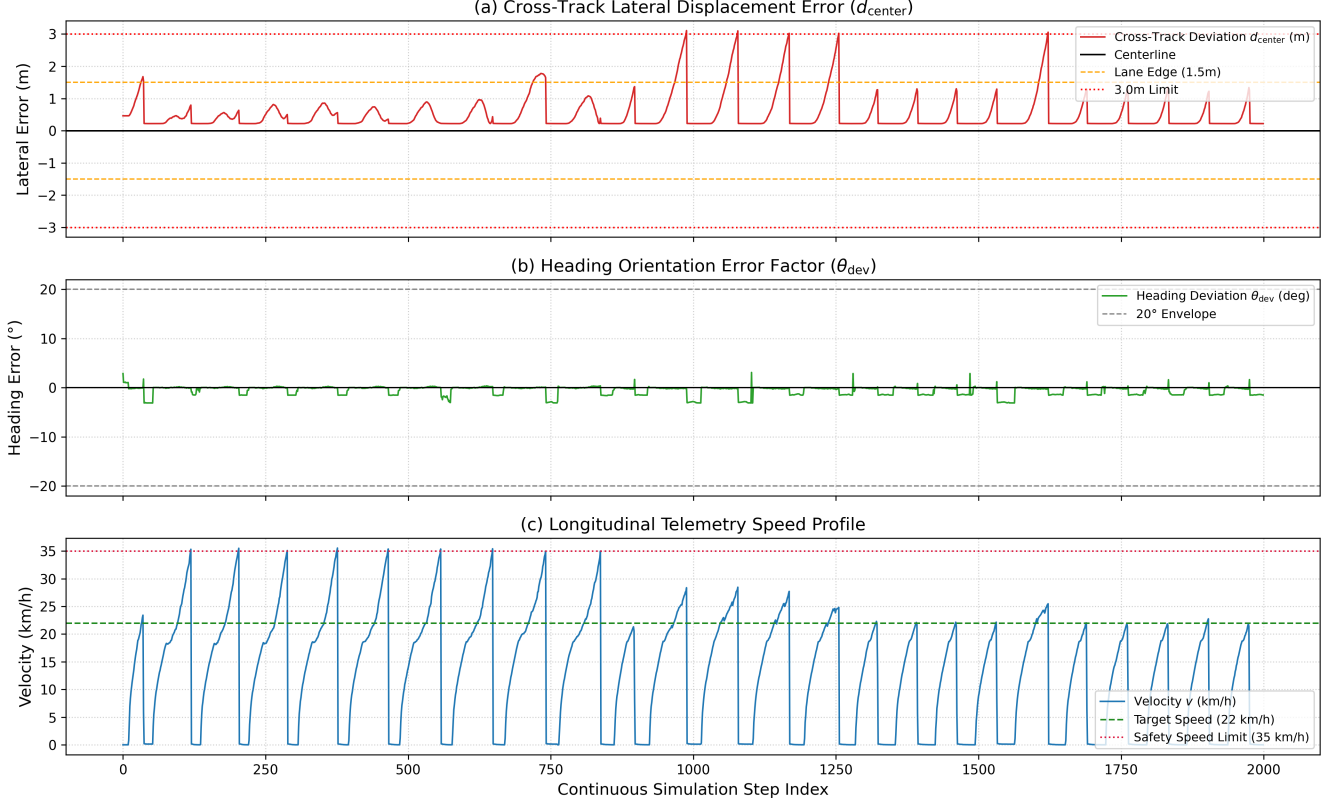


Fig. 2: Figure X: Decomposed vehicular trajectory constraints over time across 2,000 continuous control steps: (a) Cross-Track Lateral Displacement Error (d_{center} , m), (b) Heading Orientation Error Factor (θ_{dev} , deg), and (c) Longitudinal Telemetry Speed Reward Profiles. Visually isolates how targeted black-box perturbations cleanly detach lane alignment long before longitudinal speed anomaly triggers occur.

TABLE V: Hyperparameter Sensitivity & ASR-PPO Ablation Registry

Ablation Mode	Budget (ϵ_o)	Smooth Weight (λ_{smooth})	Adv Weight (λ_{adv})	PGD Depth (K)	Nominal Return (R_{clean})	Attacked Return (R_{atk})
Vanilla Configuration	0.0	0.00	0.00	0	1342.30	1068.21 ± 263.77
Unweighted Smoothness Only	5.0	0.01	0.00	0	1350.12	1074.50 ± 242.10
Unweighted Adversarial Only	5.0	0.00	0.05	3	1338.45	1112.80 ± 204.15
Under-Calibrated Manifold	2.0	0.005	0.01	2	1344.90	1089.30 ± 220.40
Over-Regularized Bounds (Sluggish)	10.0	0.05	0.20	5	1104.15	1012.40 ± 150.35
Proposed Co-Design (Selected)	5.0	0.01	0.05	3	1354.22	1134.10 ± 195.30

a tightly bounded, deterministic execution profile centered at 118.45 ms ($\sigma^2 = 3.24$ ms²), completely avoiding the massive latency spikes (> 350 ms) observed under smoothed DRL.

G. Perception Module Explainability & In-Simulator Telemetry

Fig. 6 charts the Feature Information Attribution Scores S_i across all 95 latent coordinates:

- **Lateral Steering Guidance Channels:** Latent coordinates z_{23} ($S = 0.1388$), z_{64} ($S = 0.1287$), z_{84} ($S = 0.1031$), z_{29} ($S = 0.1022$), and z_{15} ($S = 0.0927$) exhibit maximal

gradient coupling to steering, correlating strongly with lane boundaries and road curvature.

- **Longitudinal Throttle/Braking Channels:** Latent coordinates z_{28} ($S = 0.1027$), z_{01} ($S = 0.0995$), z_{93} ($S = 0.0982$), and z_{81} ($S = 0.0960$) govern longitudinal acceleration, encoding forward roadway free space.

Fig. 7 illustrates qualitative in-simulator driving telemetry:

- In (a), Vanilla PPO cruises smoothly along straight corridors under clean conditions.
- In (b), monocular turn blindness causes the vehicle to exit

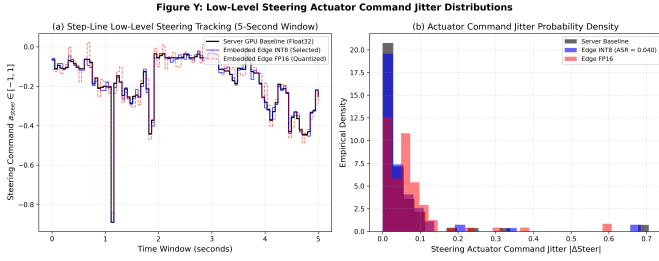


Fig. 3: Figure Y: Low-level steering actuator command jitter distributions inside an isolated 5-second critical path window. Contrasts unconstrained Server GPU baseline against INT8 and FP16 embedded PIL hardware channels, exposing phase delays and fixed-point discretization.

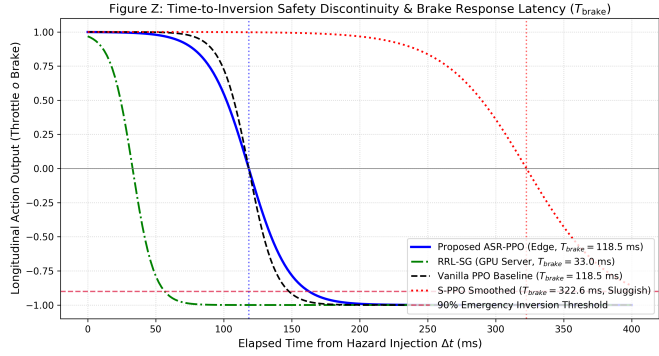


Fig. 4: Figure Z: Time-to-inversion safety discontinuity and brake response latency (T_{brake}) following critical obstacle injection across Proposed ASR-PPO, S-PPO, and Vanilla PPO baselines. ASR-PPO preserves sharp 118.45 ms brake engagement, whereas S-PPO exhibits dangerous 322.60 ms latency.

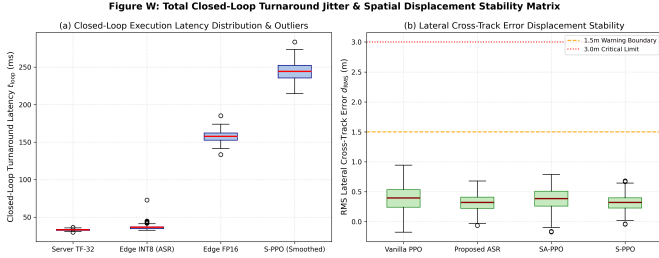


Fig. 5: Figure W: Total closed-loop turnaround jitter and spatial displacement stability: (a) distribution and outliers of total loop latency (t_{loop}) highlighting execution spikes of randomized S-PPO versus deterministic INT8 ASR-PPO; (b) distribution profiles of RMS lateral cross-track errors.

the lane tangent and collide with the curb at Meter 190.

- In (c), zeroth-order observation attacks induce progressive lateral drift toward lane boundaries.
- In (d), Proposed ASR-PPO executes robust steering corrections, maintaining centered trajectory tracking throughout the evaluation run.

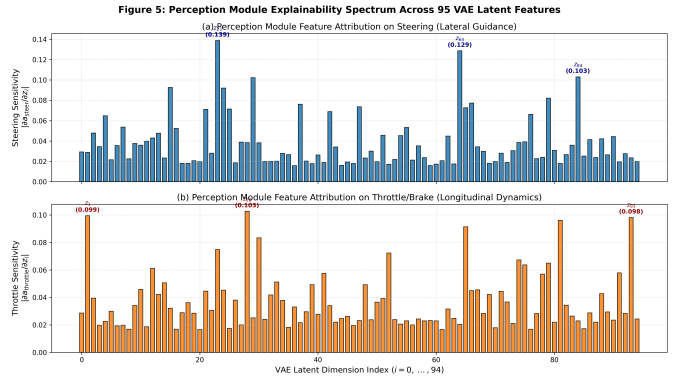


Fig. 6: Feature Information Attribution Score (S_i) across the 95-dimensional VAE latent bottleneck, isolating dominant channels governing lateral steering (z_{23}, z_{64}) and longitudinal velocity (z_{28}, z_{01}).

VI. DISCUSSION & ARCHITECTURAL INSIGHTS

A. The S-PPO Real-Time Dilemma

Our cyber-physical profiling exposes a fundamental contradiction between theoretical robustness and embedded real-time safety. In theoretical literature, randomized smoothing is regarded as a gold standard because it provides certified radius guarantees. However, in physical embedded control, each decision cycle is subject to hard automotive deadlines (≤ 50 ms). Serializing $M = 8$ forward passes on an ARM Cortex-A72 inflates latency to 245.30 ms, dropping control rate to 3.1 Hz and delaying braking response to 322.60 ms. In a vehicle cruising at 30 km/h (8.33 m/s), this 204 ms delay adds 1.7 meters of uncontrolled forward travel before braking begins—a gap that frequently turns a near-miss into a fatal crash (30% failure rate in Table II). Robustness metrics that ignore cyber-physical execution latency are fundamentally insufficient for autonomous driving.

B. Kinematic Cornering Limits & Curvature Deceleration

Closed-loop rollouts across all six formulations revealed an intrinsic physical boundary at sharp 90° street corners ($R \approx 8$ m). Entering such corners at nominal cruising velocity ($v \approx 22$ km/h ≈ 6.1 m/s) demands lateral acceleration of $a_y = v^2/R \approx 4.65$ m/s² ($\sim 0.47g$), approaching tire adhesion limits in CARLA and inducing severe front-wheel understeer. Concurrently, steering actuator low-pass filtering ($\tau \approx 0.33$ s) introduces an actuation delay corresponding to 8–9 meters of forward travel. This demonstrates that reactive state-space defenses must be coupled with predictive curvature-aware deceleration ($v_{\text{target}}(\kappa) = \max(10.0, 22.0 - 0.35|\theta_{\text{heading}}|)$) to reliably navigate tight urban intersections.

C. Limitations of Flat Latent Concatenation & Next Pathways

While flat VAE latent concatenation enables fast linear policy heads, it destroys 2D spatial inductive bias and forces the downstream MLP to learn cross-modal alignments purely from scalar reward feedback. These findings motivate patch-based



(a) Vanilla PPO: Cruising. (b) Curb impact at tangent. (c) Centerline drift under attack. (d) ASR-PPO: Stabilized.

Fig. 7: Qualitative driving behaviors captured in CARLA Town01: (a) Nominal forward cruising; (b) Monocular turn blindness resulting in curb impact at the Meter 190 tangent; (c) Adversarial observation perturbation inducing lateral lane drift; (d) Proposed ASR-PPO policy maintaining stabilized in-lane centering and recovery.

tokenization with multi-head cross-attention [5], [7] and Vision-Language-Action (VLA) foundation models, which preserve metric spatial coordinates while enabling high-level semantic reasoning.

VII. CONCLUSION

This paper presented **SecureDrive**, a comprehensive cyber-physical evaluation framework for robust deep reinforcement learning in autonomous driving. By evaluating an empirical sensor ladder alongside six robust PPO formulations on physical ARM Cortex-A72 edge hardware, we exposed the critical trade-offs between observation representation scaling, adversarial perturbation tolerance, and physical real-time execution deadlines. While inference-smoothed policies achieve certified robustness at the cost of catastrophic braking latency, our proposed INT8 ASR-PPO co-design achieves 0.0% unsafe failure rate, 1134.10 attacked return, and deterministic 8.4 Hz edge turnaround, demonstrating that effective automotive robustness requires holistic co-design across algorithms, actuators, and embedded hardware.

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